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import random
```

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# This code was developed by the student with AI-assisted  
debugging and refinement.
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```
def analyze_guess(current_guess, secret_number, guess_history):  
    """  
    Analyzes the player's guess using sequencing, selection, and  
iteration.  
    Parameters:  
        current_guess (int): User's current guess  
        secret_number (int): Random target number  
        guess_history (list): List of all guesses  
    Returns:  
        str: Result message  
    """
```

```
# Iteration through guess history to calculate total  
attempts
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```
    attempt_count = 0  
    for guess in guess_history:  
        attempt_count += 1
```

```
# Selection logic
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    if current_guess > secret_number:  
        return f"Too high! Attempts so far: {attempt_count}"  
    elif current_guess < secret_number:  
        return f"Too low! Attempts so far: {attempt_count}"  
    else:  
        return f"Correct! You guessed it in {attempt_count}  
attempts."
```

```
# Main game loop
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```
play_again = True
```

```
while play_again:  
    secret_number = random.randint(1, 100)  
    guesses = []  
    guessed_correctly = False
```

```
    print("\nWelcome to the Number Guessing Game!")  
    print("I am thinking of a number between 1 and 100.")
```

```
    while not guessed_correctly:  
        try:
```

```

        user_guess = int(input("Enter your guess: "))

        # Input validation
        if user_guess < 1 or user_guess > 100:
            print("Please enter a number between 1 and
100.")
            continue

        # Store guess in list
        guesses.append(user_guess)

        # Call student-developed procedure
        result = analyze_guess(user_guess, secret_number,
guesses)
        print(result)

        # End game if correct
        if user_guess == secret_number:
            guessed_correctly = True

    except ValueError:
        print("Invalid input. Please enter a whole number.")

    # Output full guess history
    print("Your guesses were:", guesses)

    # Additional list usage to manage complexity
    print("Highest guess:", max(guesses))
    print("Lowest guess:", min(guesses))
    print("Average guess:", round(sum(guesses) / len(guesses),
2))

    # Replay option
    replay = input("Would you like to play again? (yes/no):
").lower()

    if replay != "yes":
        play_again = False
        print("Thanks for playing!")

```